Breast Cancer Detection Using Support Vector Machine

A Study on a Custom Synthetic Dataset of Jharkhand

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Machine Learning (AI26509)

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Introduction & Problem Statement

- **Healthcare Challenge:** Breast cancer classification is an important area of healthcare research. Early identification can support timely medical evaluation.
- **Machine Learning:** Machine learning can study patterns in data and classify records into different categories.
- **Our Approach:** We use Support Vector Machine (SVM) to classify tumor records as Benign or Malignant.
- **Regional Focus:** The project uses a custom synthetic dataset representing six regions of Jharkhand.

Problem Statement

Our objective is to build a classification model that can distinguish between Benign and Malignant records using five selected features.

Research Objectives & Related Studies

Research Objectives

1. Use a synthetic dataset containing 569 records.
2. Implement SVM with an RBF kernel.
3. Prepare and scale the input features.
4. Compare SVM with a Decision Tree model.
5. Evaluate the models using standard metrics.

Related Studies

- Zhou et al. (2022): SVM-based analysis of electronic medical records.
- Abdelfattah et al. (2023): Privacy-preserving medical diagnosis using SVM.
- Guido (2024): Overview of SVM methods and kernel techniques.
- Huynh et al. (2024): SVM application in machinery fault detection.

Understanding Support Vector Machine

What is SVM?

SVM is a machine learning algorithm that finds a boundary to separate two or more classes.
For our project, the two classes are:

Benign

A record belonging to the Benign category.

Malignant

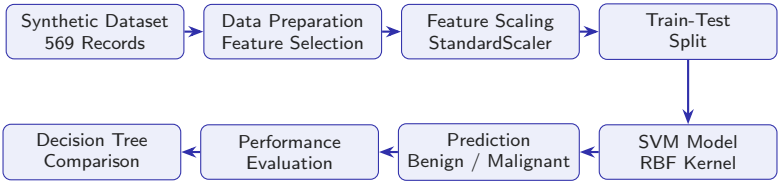
A record belonging to the Malignant category.

Important SVM Concepts

- **Decision Boundary:** The line or surface that separates classes.
- **Support Vectors:** The closest data points that help define the decision boundary.
- **RBF Kernel:** Helps SVM create a curved boundary when a straight line is not enough.

Proposed System Architecture

How the project works from data to prediction



Tools Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib and Seaborn.

Dataset Description & Distribution

Dataset Details

- 569 Records
- 6 Regions
- 5 Input Features
- 2 Diagnosis Classes
- 2023–2026 Dataset Period

Diagnosis Count

- Benign: 351
- Malignant: 218

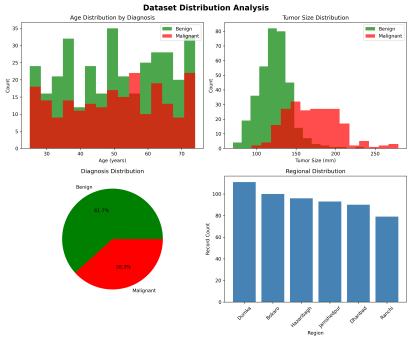


Figure 1: Distribution of records across diagnosis, features and regions.

Methodology

Step-by-Step Process

1. **Load Dataset:** Read the CSV file containing 569 records.
2. **Select Features:** Use Age, Tumor Size, Lymph Node Status, Family History and BMI.
3. **Encode Diagnosis:** Benign is represented as 0 and Malignant as 1.
4. **Split the Dataset:** Use 80% records for training and 20% for testing.
5. **Scale Features:** StandardScaler brings the input features to a common scale.
6. **Train Models:** Train SVM and Decision Tree using the same train-test split.
7. **Evaluate Results:** Compare accuracy, precision, recall, F1-score and confusion matrix.

Python Implementation

Core SVM Code

```
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

svm_model = SVC(
    kernel='rbf',
    C=1.0,
    gamma='scale')

svm_model.fit(X_train_scaled, y_train)
y_pred = svm_model.predict(X_test_scaled)
```

Implementation Summary

The model learns from the training records and predicts the diagnosis of unseen test records.

Experimental Results: Confusion Matrix

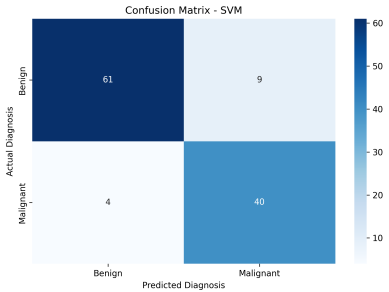


Figure 2: Confusion Matrix of the SVM model.

Observed Values

- True Negative: 61
- False Positive: 9
- False Negative: 4
- True Positive: 40

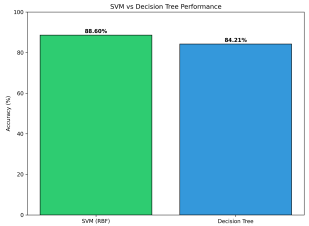
Important Observation

The model incorrectly classified 4 Malignant records as Benign.

Correct Predictions: 101 out of 114

SVM vs Decision Tree

Accuracy Comparison



Algorithm	Accuracy
SVM (RBF)	**88.60%**
Decision Tree	84.21%

Observation

SVM achieved 4.39 percentage points higher accuracy than the Decision Tree baseline on this test split.

Other Results and Visualizations

Feature Correlation

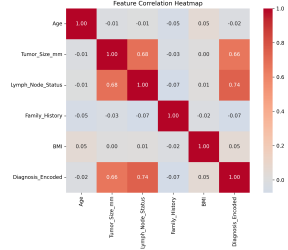


Figure 4: Correlation between selected features and diagnosis.

SVM Decision Boundary

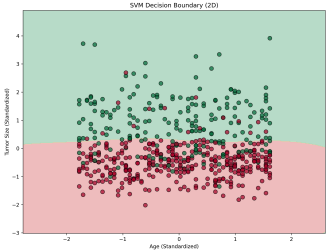


Figure 5: Two-feature visualization using Age and Tumor Size.

The decision boundary is a separate 2D visualization. The main SVM model uses all five input features.

Detailed SVM Performance

SVM Evaluation Results

Evaluation Metric	SVM (RBF)
Accuracy	88.60%
Precision	81.63%
Recall	90.91%
F1-Score	86.02%

What Do These Results Mean?

- **Accuracy:** Overall percentage of correct predictions.
- **Precision:** How many predicted Malignant records were actually Malignant.
- **Recall:** How many actual Malignant records were correctly identified.
- **F1-Score:** A combined measure of precision and recall.

Limitations & Future Work

Limitations

- Uses synthetic data instead of real patient records.
- Uses only five input features.
- No clinical validation has been performed.
- Current implementation is a local Python prototype.

Future Work

- Collect real patient data with appropriate ethical approval.
- Include more relevant features.
- Develop a secure web application.
- Test the model on independent datasets.

Future Direction

Move from synthetic experiments towards testing with real-world data.

Conclusion

Project Summary

This project presents an SVM-based approach for classifying synthetic breast tumor records.

- Used a dataset containing 569 synthetic records.
- Selected five input features for classification.
- Applied feature scaling before SVM training.
- Implemented SVM with an RBF kernel.
- Compared SVM with a Decision Tree baseline.
- Achieved 88.60% SVM accuracy and 84.21% Decision Tree accuracy.

Important Note

This is an academic machine learning experiment, not a clinically validated diagnostic system.

References

References

1. Zhou et al. (2022). SVM-based analysis of electronic medical records.
2. Abdelfattah et al. (2023). Privacy-preserving medical diagnosis using SVM.
3. Guido (2024). Overview of SVM methods and kernel techniques.
4. Huynh et al. (2024). SVM application in machinery fault detection.

Thank You!

Questions & Discussion

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